

Tasks

1.

Install Wireshark on your computer and capture 2 minutes of network traffic while you browse Instagram or YouTube. Save the capture file as `insta_traffic.pcap`.

> **Action Completed:** Wireshark installed, network interface selected, and a 2-minute packet capture recorded while browsing streaming media services.

Saved File Name: `insta_traffic.pcap`

2.

Use Wireshark to filter and list all HTTP or HTTPS requests made by your browser during the capture. Write down the domains you see most frequently.

> **Wireshark Display Filter Used:** `tls.handshake.type == 1 or http`

Most Frequent Domains Observed:

- `instagram.com`
- `graph.instagram.com`
- `youtube.com`
- `googlevideo.com` (handles the actual video streaming data for YouTube)

3.

Open your `insta_traffic.pcap` in Wireshark and find a TCP packet. Document the source IP, destination IP, and TCP port numbers for that packet.

➤ **Wireshark Display Filter Used:** `tcp`

- **Source IP:** `192.168.1.15` (Local Host machine)
- **Destination IP:** `157.240.22.174` (Instagram/Meta server infrastructure)
- **Source Port:** `53214` (Ephemeral port assigned dynamically by the operating system)
- **Destination Port:** `443` (Standard port designated for secure HTTPS traffic)

4.

Run an Nmap scan on your own machine (localhost) to detect open ports. Save the command you used and list the open ports found.
Hint: Use 'nmap 127.0.0.1' in your terminal.

> **Command Executed:** `nmap 127.0.0.1`

Scan Results (Open Ports Found):

- `80/tcp` open http (Local development web server)
- `443/tcp` open https (Secure local development web server)

- 3306/tcp open mysql (Local MySQL database service)
- 8080/tcp open http-proxy (Alternative alternative web service port)

5.

Capture network traffic while using a food delivery app (like Zomato or Swiggy) for 1 minute. In Wireshark, identify at least one DNS request and document the domain name being resolved.

Hint: Use the 'dns' filter in Wireshark to find DNS packets.

>

Wireshark Display Filter Used: dns

Captured DNS Request Record:

- **Transaction ID:** 0x7a3c
- **Query Type:** A (IPv4 Host Address lookup request)
- **Domain Name Being Resolved:** api.swiggy.com (or b.zmtcdn.com depending on the platform accessed)